# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

**III Semester of BSc (BT/MI/BC) Examination (Backlog) March 2018**

**Course code & Name: BS305: Fundamentals of Biochemistry-II**

**Date: 04/04/2018 Day: Wednesday Time: 10:00 am To 10:20 am Maximum Marks: 10**

## MCQ

**Important Instructions:**

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

**Q - I Choose the correct answer for the following questions. 10**

**1.** A fatty acid with 16 carbon atom will undergo how many cycle of B- oxidation?

(i) 6 (ii) 7
(iii) 8 (iv) 16

**2.** Which of the following statements correctly describes the entropy of the universe, as captured by the second law of thermodynamics?

(i) The total entropy of the universe fluctuates between being high and being low.
(ii) The total entropy of the universe remains constant.
(iii) The total entropy of the universe is continually decreasing.
(iv) The total entropy of the universe is continually increasing.

**3.** A kinase is an enzyme that______________.

(i) removes phosphate groups of substrates
(ii) uses ATP to add a phosphate group to the substrate
(iii) uses NADH to change the oxidation state of the substrate
(iv) removes water from a double bond

**4.** How many ATPs are generated from one glucose molecule by substrate level phosphorylation during the glycolysis?

(i) 1 ATP (ii) 2 ATP
(iii) 4 ATP (iv) 8 ATP

**5.** Which of the following is/are fat soluble vitamin?

(i) Vitamin D (ii) Vitamin E
(iii) Vitamin A (iv) all of these

**6.** Pentose phosphate pathway is also known as ___________.

(i) Hexose mono-phosphate shunt (ii) Embden Meyerhof pathway)
(iii) Krebs pathway (iv) None of these

**7.** The precursor to glycogen in the glycogen synthase reaction is_________.

(i) glucose-1-P (ii) UDP-glucose
(iii) UTP-glucose (iv) glucose-6-P

**8.** _________is a quantitative expression for the randomness or disorder in a system.

(i) Entropy (ii) Energy
(iii) Enthalpy (iv) all of these

**9.** Reactions of the TCA cycle occur in ________.

(i) mitochondria (ii) cytoplasm
(iii) Vacuoles (iv) lysosome

**10.** Among the following, which amino acid served as nitrogen source for the synthesis of purine through de-novo synthesis?

(i) Isoleucine (ii) Aspartate
(iii) Histidine (iv) Lysine

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**III Semester of BSc (BT/MI/BC) Examination (Backlog) March 2018**

**Course code & Name: BS305: Fundamentals of Biochemistry-II**

**Date: 04/04/2018 Day: Wednesday Time: 10:20am To 12:00 pm Maximum Marks: 25**

*Instructions:*

1. **Section I and II must be attempted in SINGLE ANSWER SHEET.**
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

## SECTION –I

**Q – II Answer the following questions as directed (any five) (2 Marks X 5 ) 10**

1. Write any two differences between glycolysis and gluconeogenesis
2. Enlist the different reactions of β-Oxidation
3. What is first law of thermodynamics? Explain in brief with example.
4. Write a note on ‘ Nucleotide metabolic disorders’
5. Write a note on “Water soluble vitamins”.
6. Define: Energy and Enthalpy
7. What is Gibbe’s free energy? When the value of $\Delta G=0$, $\Delta G>0$ and $\Delta G<0$ in the reaction?

## SECTION – II

**Q-III Answer the following questions as directed (any three) (5 Marks X 3 ) 15**

1. Explain in detail about digestion, mobilization, and transport of fats.
2. What id TCA cycle? Explain in detail about different reaction steps of TCA cycle.
3. Explain the second law of thermodynamics with an appropriate example.
4. What is importance of urea cycle? Describe various steps of urea cycle.